

# Situation-Aware Integrated Sensing and Communication for 6G Smart Warehouses: A Sionna RT and ns-3 Co-Simulation Framework

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**Abstract**—The evolution toward 6G is driven by stringent latency, reliability, and throughput requirements of Digital-Twin (DT) applications such as smart warehouses, autonomous systems, and intelligent environments. In these settings, the wireless network is no longer a passive medium but an active component that must adapt continuously to user position, mobility, and environmental geometry. This motivates situation-aware operation, where the antenna system anticipates rather than reacts to channel conditions. This paper presents a situation-aware Integrated Sensing and Communication (ISAC) framework that couples NVIDIA Sionna RT for GPU-accelerated, site-specific ray tracing with ns-3/5G LENA for full-stack NR networking. The framework is organised into four modules: a radar-inspired sensing module that converts ray-traced multipath into Equivalent Reflection Points (ERPs) and applies *Z*-gating, Moving-Target Indication (MTI), and DBSCAN clustering; a multi-target tracking module that combines a constant-velocity Kalman filter with Hungarian data association; a sensing-informed adaptive beamforming module that synthesises a multi-lobe Gaussian radiation pattern toward confirmed tracks; and a communication module that exports per-path channel state to ns-3. The framework is validated in two stages. A controlled stage exercises the pipeline in free-space, warehouse, and urban-micro street-canyon scenes with six ablation variants. An application-oriented stage embeds the framework in an end-to-end smart-warehouse Digital Twin with mixed mobile-robot, sensor, and video traffic, evaluated for  $2 \times 2$ ,  $4 \times 4$ , and  $8 \times 8$  gNB arrays. Results show up to 18 dB reduction in mean path loss in free space, detection rates above 0.84, and—in the warehouse use case—a  $4 \times 4$  array as the balanced operating point, with roughly 25 dB reduction in robot-link path loss and improved aggregated throughput at near-baseline energy cost. The proposed framework is therefore viable for deployment-oriented ISAC studies once sufficient gNB array support is available.

**Index Terms**—Integrated sensing and communication (ISAC), 6G, situation awareness, ray tracing, Sionna, ns-3, 5G LENA, adaptive beamforming, multi-target tracking, Kalman filter, Digital Twin, smart warehouse.

## I. INTRODUCTION

The evolution toward 6G networks is driven by the need to support increasingly complex and dynamic applications, including Digital Twins (DT), autonomous systems, and intelligent environments. Among these, smart-warehouse systems represent a particularly challenging use case, where mobile robots, sensing devices, and control systems must operate reliably under stringent latency, reliability, and throughput requirements. In such environments, the wireless network is no longer a passive communication medium: it becomes an active

component of the system that must adapt continuously to changes in user position, mobility patterns, and environmental conditions. This motivates the concept of *situation awareness*, in which the communication system leverages contextual information to optimise its behaviour proactively rather than reactively [1].

Situation awareness, in the context of wireless access networks, refers to the ability of the network—particularly the antenna system and its associated control logic—to perceive and exploit contextual information such as user equipment (UE) location, mobility trajectories, surrounding geometry, and potential obstacles. Instead of relying on periodic beam training or threshold-based handovers, a situation-aware system anticipates future channel conditions and adapts accordingly. In a smart-warehouse setting, this includes predicting robot movement along planned paths, identifying potential blockages from racks or moving objects, and adjusting beamforming strategies to maintain reliable links. Such proactive adaptation has been shown to improve key performance indicators (KPIs) including latency, throughput, reliability, and energy efficiency, particularly in higher-frequency bands where propagation is highly sensitive to environmental dynamics [2], [3].

Enabling situation awareness, however, requires the network to have access to accurate and timely environmental information. Conventional radio measurements such as RSRP or SINR provide limited insight into the physical environment and often arrive too late to support proactive adaptation. To overcome this limitation, the antenna system must be endowed with *sensing capabilities*, allowing it to observe the surrounding environment and infer the position and motion of objects directly. This convergence of sensing and communication functionalities is a fundamental requirement for next-generation networks and forms the basis of the Integrated Sensing and Communication (ISAC) paradigm. ISAC integrates radar-like sensing into the communication infrastructure so that the same hardware, waveform, and spectrum are jointly used for data transmission and environmental sensing. In a smart warehouse, ISAC enables a base station to detect and track mobile robots and other dynamic objects from reflected signals, transforming the communication system into a distributed sensing platform without dedicated sensing infrastructure.

A key challenge in studying ISAC systems lies in faithfully modelling the interaction between electromagnetic waves and complex environments. Conventional system-level simulators

rely on stochastic channel models that do not capture site-specific reflections, diffractions, and blockages. To address this, ray-tracing techniques are employed as a high-fidelity modelling approach: by leveraging detailed scene geometry and material properties, ray tracing provides per-path delays, angles of arrival/departure, and complex gains, which are essential both for accurate communication modelling and for the radar-like reasoning that underpins sensing [4], [5].

Building on this foundation, this paper presents an ISAC framework that pairs NVIDIA Sionna RT with ns-3/5G LENA into a single co-simulation pipeline, and evaluates it both in controlled scenes and in an end-to-end smart-warehouse Digital Twin. The contributions of the paper are as follows:

- It describes the integration of ns-3/5G LENA with Sionna RT through a ZeroMQ-based propagation/sensing bridge, with a clean role boundary between the two simulators (Section III).
- It presents a four-module ISAC architecture comprising a radar-inspired sensing module (Equivalent Reflection Point (ERP) computation,  $Z$ -gating, Moving-Target Indication (MTI), and DBSCAN clustering), a Kalman–Hungarian multi-target tracking module, a sensing-informed multi-lobe adaptive beamforming module, and a communication module that exports per-path channel state back to ns-3 (Section III).
- It validates the framework through a two-stage evaluation: a controlled scene-level study across free-space, warehouse, and urban-micro street-canyon scenes with six ablation variants, and an application-oriented warehouse Digital Twin study under three gNB array sizes (Section IV).
- It identifies the operating regimes in which the proposed framework is beneficial, including the  $4 \times 4$  gNB configuration as the balanced operating point in the smart-warehouse scenario, and discusses the remaining limitations and future research directions (Sections IV–V).

The remainder of this paper is organised as follows. Section III describes the simulation flow, ISAC architecture, and the algorithms used in each module. Section IV presents the controlled-validation and warehouse-use-case results together with a discussion of their implications. Section V concludes the paper and outlines future work.

## II. BACKGROUND AND RELATED WORK

This section first summarises the gains that situation awareness can bring to next-generation access networks, and then discusses the simulation tooling that is required to study these gains in a site-specific manner. The latter discussion directly motivates the choice of ns-3/5G LENA combined with Sionna RT adopted in this paper.

### A. Benefits of Situation Awareness

Situation awareness refers to the capability of antenna systems to adapt their configuration based on contextual information such as UE position, mobility, environment geometry, obstacles, and temporal channel variation. Recent studies show

that incorporating such awareness leads to substantial gains in performance, reliability, and efficiency, which can be grouped into four families.

1) *Beamforming and Beam Management*: Surveys of AI-enabled multimodal beamforming indicate that combining non-RF sensors (e.g., LiDAR, radar, GPS) with RF data improves directional accuracy and reduces the overhead of beam sweeping and alignment, particularly in mmWave bands where misalignment and rapid channel variation degrade performance [2]. Positioning-aided beamforming for mmWave communications has specifically shown how location information can relax the complexity of beam training and narrow the search space for optimal beams.

2) *Mobility and Handover Management*: Context-aware and predictive handover schemes that exploit UE speed, direction, hysteresis, trajectory history, and signal-quality metrics reduce handover failures, ping-pong effects, and latency. Memory-full context-aware predictive mobility management in dual-connectivity 5G networks uses past UE trajectory and signal measurements to anticipate handover events and reduce signalling overhead [3]. Surveys on AI-aided mobility and handover management confirm that machine-learning-based schemes can sustain seamless connectivity even in ultra-dense deployments and high-mobility regimes.

3) *Reliability and Link Stability*: Awareness of obstacles and environmental geometry enables dynamic beam switching or alternate-path selection, mitigating the impact of blockages especially in higher-frequency bands. Future networks are expected to rely heavily on integrated sensing (and positioning) capabilities to improve link robustness under dynamic conditions [1].

4) *Resource Efficiency and Latency Reduction*: Predictive mobility and proactive beam adjustment avoid unnecessary sweeping, measurement reporting, and control overhead. Deep-learning-based predictive handover algorithms have been shown to reduce interruption time and improve resource utilisation; in mobile edge-computing systems, joint optimisation of connectivity and computation likewise benefits from context-aware decision rules [3].

While the literature confirms substantial improvement potential, it also emphasises trade-offs: acquiring accurate context (position, environment mapping) has its own overhead, prediction errors can degrade rather than improve performance, and privacy and hardware constraints must be addressed.

### B. Simulation Tooling for ISAC

Studying situation-aware ISAC requires a simulation environment that captures both end-to-end networking behaviour and site-specific propagation. Two simulator families are commonly used as starting points.

**Simu5G on OMNeT++**. Simu5G provides end-to-end simulation of 5G NR spanning application, transport, and network layers, with support for MEC, C-V2X, and dual connectivity. However, its PHY relies on stochastic 3GPP abstractions, so it lacks site-specific 3D channel modelling and flexible MIMO configurations. Most critically for ISAC, the OMNeT++ architecture makes integrating GPU-accelerated ray tracers such as

Sionna RT onerous: the co-simulation bridge requires heavy glue code, Simu5G exposes only omnidirectional antennas, and the rich channel outputs computed in Sionna (per-path gains, delays, AoA/AoD) cannot be propagated cleanly to the upper layers.

**5G LENA on ns-3.** 5G LENA implements PHY/MAC system-level behaviour aligned with 3GPP standards and supports full-stack studies, from packet capture and IP transport up to RAN-layer scheduling and application flows [6]. By default it uses simplified stochastic channel models (e.g., 3GPP TR 38.901 [7]) for path loss, fading, and beamforming, and therefore lacks deterministic 3D geometry awareness. However, recent work on Ns3Sionna [5] shows that Sionna RT’s GPU-accelerated ray tracing can be integrated directly into ns-3, yielding high-fidelity site-specific channels within the existing ns-3 stack; the published Wi-Fi integration serves as a proof of concept and a solid technical foundation to extend the same approach to 5G LENA. In this ns-3/LENA path, integration is cleaner and more expressive: both sides expose MIMO/beamforming configuration, the per-path channel state computed in Sionna (complex gains, delays, Doppler, AoA/AoD) can be passed through intact, and the resulting scheduling, link adaptation, and application-layer metrics faithfully reflect the underlying geometry-driven radio conditions.

For these reasons, the framework presented in this paper pairs 5G LENA with Sionna RT [4] and gives each tool a clear role: Sionna handles the physical and link layers (ray tracing, MIMO arrays/beamforming, per-path/beam loss and delay), while 5G LENA consumes these geometry-aware results to drive the upper layers (RLC/PDCP/IP/applications), so that buffering, congestion control, and application QoE evolve in response to the same radio conditions produced by Sionna. This division of labour is detailed in Section III.

### III. FRAMEWORK AND METHODOLOGY

This section describes the proposed situation-aware ISAC framework. We first present the smart-warehouse use case that motivates the design, then the co-simulation architecture and flow, and finally the four functional modules of the ISAC pipeline together with their underlying algorithms.

#### A. Smart-Warehouse Use Case

The chosen reference scenario is a logistics floor served by a single gNB and populated by heterogeneous wireless devices: stationary robotic arms along an intake conveyor, autonomous mobile robots that ferry parcels to and from storage racks, fixed cameras, environmental sensors (temperature, humidity), package and rack sensors, and PCs in a control room running monitoring applications (Fig. 1). The floor exhibits three RF regions—a clear line-of-sight zone, a partially obstructed zone with rack edges that intermittently shadow links, and a fully obstructed zone (control room) separated by walls and dense shelving—which exercise the full range of LOS/NLOS, diffraction, and blockage conditions.

This geometry exercises three progressively more demanding scenarios: a stationary baseline, predefined mobile paths

Fig. 1. Smart-warehouse Digital Twin: a single gNB serves stationary robots, autonomous mobile robots, sensors, and fixed cameras across three RF regions.

through mixed-obstruction zones, and dynamic beamforming with position feedback. The scenario also represents the broader class of situation-aware use cases identified in our prior work (smart home, smart warehouse, vehicular networks): all three share the need for proactive beam steering, predictive handover, and blockage-aware scheduling. The smart-warehouse case is selected here because it cleanly combines device heterogeneity, mobility-driven channel variation, and a controlled indoor geometry suitable for ray-traced sensing.

#### B. Co-Simulation Architecture

The end-to-end architecture is shown conceptually in Fig. 2. A remote host runs the application over TCP/UDP/IP and attaches to the EPC via the PGW/SGW node (EpcPgwSgwApp over GTP/UDP/IP). On the RAN side, the gNB executes EpcEnbApp and the LENA NR stack: packets from the core arrive in GTP/UDP/IP, are scheduled by the NR MAC/PHY, and are forwarded to the gNB’s `NrGnbNetDevice` for transmission over the air. The UE mirrors this stack (application → TCP/UDP → IP → `NrUeNetDevice`), so end-to-end KPIs (throughput, latency, jitter, loss) are measured natively within ns-3.

Radio propagation is outsourced to a dedicated Sionna “channel server” co-located with the RAN. Given the true 3D scene and configured MIMO arrays, Sionna computes time-varying path sets (per-path gains, delays, Doppler, AoA/AoD) and returns effective propagation loss and delay—and, when needed, per-RB SINR estimates—through a ZeroMQ-based adapter. The gNB therefore performs link adaptation and scheduling against geometry-aware channels that reflect LOS/NLOS transitions, diffraction, blockage, and beamforming, while the upper layers react naturally to the resulting radio dynamics.

#### C. Simulation Flow

In ns-3 we first define the gNB and UE nodes, attach mobility models and antenna/array configurations, and select the application mix (e.g., constant-bit-rate, VoIP, video streaming). The initial positions, MIMO settings, and application

Fig. 2. End-to-end co-simulation architecture. Sionna RT supplies geometry-aware physical-layer realism; ns-3/5G LENA consumes it to drive the protocol stack and applications.

parameters are sent to the Sionna server. Sionna has already created the 3D scene and an occupancy map, and waits for requests; upon receiving ns-3's parameters, it initialises the ray-tracing environment, sets Tx/Rx poses and MIMO patterns, and enters a request/response loop. Whenever ns-3 asks for channel state, Sionna updates node poses, estimates paths via ray tracing, optionally adapts the transmit radiation pattern to the current UE positions, and returns time-aligned propagation loss and delay reflecting LOS/NLOS transitions, multipath, diffraction, and blockage. ns-3 ingests these values at the PHY/MAC, performs link adaptation and scheduling, and the upper layers evolve naturally over those channels.

#### D. Scene and Mobility Modelling

The 3D scene is loaded from a Blender-exported Mitsuba XML file and instantiated three times in Sionna RT,  $\mathcal{S} = \{\mathcal{S}_{\text{sense}}, \mathcal{S}_{\text{comm}}, \mathcal{S}_{\text{bf}}\}$ . The three scenes share geometry but differ in runtime use:  $\mathcal{S}_{\text{sense}}$  uses a co-located transmit and receive array as a monostatic radar;  $\mathcal{S}_{\text{comm}}$  uses the baseline transmitter to compute per-link channel state between the gNB and UEs; and  $\mathcal{S}_{\text{bf}}$  uses a custom multi-lobe radiation pattern computed from the sensing output. Maintaining three scene instances avoids repeatedly mutating a single Sionna object between modes.

Mobility is handled by a headless PyBullet backend. Static geometry is loaded as a mesh, while dynamic objects are loaded as box collision bodies derived from the bounding box of each mesh. For an object whose mesh occupies  $\mathbf{v}_{\min} = [x_{\min}, y_{\min}, z_{\min}]^T$  and  $\mathbf{v}_{\max} = [x_{\max}, y_{\max}, z_{\max}]^T$ , the half-extents used for collision modelling are  $\mathbf{h} = \frac{1}{2}(\mathbf{v}_{\max} - \mathbf{v}_{\min})$ . Because PyBullet operates on the geometric centre while users specify positions at the base, the centre position is recovered as  $\mathbf{p}_{\text{centre}} = \mathbf{p}_{\text{bottom}} + [0, 0, h_z]^T$ .

Path planning uses the A\* algorithm on a 2D grid of resolution  $g$ . Each cell  $n$  is scored by  $f(n) = g(n) + h(n)$ , with  $g(n)$  the cost-so-far (straight moves cost  $g$ , diagonals cost  $g\sqrt{2}$ ) and  $h(n) = \sqrt{(x_n - x_d)^2 + (y_n - y_d)^2}$  the Euclidean heuristic to the destination. Candidate paths are validated against static and dynamic collisions before commitment. Once accepted, a path is stored as an ordered list of waypoints

Fig. 3. ISAC pipeline: sensing, multi-target tracking, adaptive beamforming, and communication.

$\{w_0, \dots, w_K\}$ , and at simulation time  $t \geq t_0$  the agent's traversed length is  $s(t) = v(t - t_0)$ . Defining  $D_0 = 0$  and  $D_k = \sum_{i=1}^k \|w_i - w_{i-1}\|_2$ , the instantaneous position on segment  $[w_k, w_{k+1}]$  when  $D_k \leq s(t) \leq D_{k+1}$  is obtained by linear interpolation,

$$\mathbf{p}(t) = w_k + \alpha(w_{k+1} - w_k), \quad \alpha = \frac{s(t) - D_k}{D_{k+1} - D_k}. \quad (1)$$

#### E. ISAC Pipeline Overview

The ISAC pipeline (Fig. 3) consists of four modules executed in sequence at every sensing tick: a sensing module that converts raw ray-tracing output into 3D detections, a multi-target tracking module that stabilises detections into trajectories, an adaptive beamforming module that synthesises a custom transmit pattern from the confirmed tracks, and a communication module that re-runs ray tracing under the new pattern to produce the per-link channel state delivered to ns-3.

#### F. Ray-Tracing-Based Sensing Module

The sensing module transforms the ray-traced multipath output into discrete 3D object candidates, following a radar-like reasoning.

a) *Pre-processing.*: For each path  $\ell$ , the average path power is computed from the complex coefficients,  $\overline{P}_\ell = \frac{1}{|I|} \sum_{i \in I} |a_{\ell,i}|^2$ , together with the average delay  $\overline{\tau}_\ell = \frac{1}{|I|} \sum_{i \in I} |\tau_{\ell,i}|$  and the average angles  $\overline{\theta}_\ell$  and  $\overline{\phi}_\ell$ . Paths with  $\overline{\tau}_\ell \leq 10^{-9}$  s are dropped. The remaining delays are converted to range,  $L_\ell = c\overline{\tau}_\ell$ . A distance-adaptive power threshold avoids over-suppressing distant returns,

$$s_\ell = \max\left(\frac{L_\ell}{10}, 1\right)^{1.1}, \quad P_{\text{th},\ell} = \frac{P_{\min}}{s_\ell}, \quad (2)$$

and only paths with  $\overline{P}_\ell > P_{\text{th},\ell}$  are retained. Finally, each surviving angle pair is converted to a 3D unit direction,  $\mathbf{d}_\ell = [\sin \theta_\ell \cos \phi_\ell, \sin \theta_\ell \sin \phi_\ell, \cos \theta_\ell]^T$ .

b) *Equivalent Reflection Point (ERP).*: Each surviving path is mapped to a single 3D point,

$$\mathbf{p}_\ell^{\text{ERP}} = \mathbf{p}_{\text{gNB}} + \left(\frac{L_\ell}{2} + \delta\right) \mathbf{d}_\ell, \quad (3)$$



where  $\mathbf{p}_{\text{gNB}}$  is the gNB location,  $L_\ell/2$  accounts for the round-trip in the monostatic case, and  $\delta$  is a small calibration offset that mitigates the bias introduced when reflection occurs on a surface rather than at the object's centre.

*c) Vertical (Z) gating.:* Floor and ceiling reflections produce ghost ERPs that lie outside the height range of physical targets. A simple gate  $z_{\min} \leq z_{\ell}^{\text{ERP}} \leq z_{\max}$ , calibrated from the target mesh, removes them.

*d) Moving-Target-Indication (MTI) filter.:* The first frame is stored as a static background point cloud  $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_M\}$ . For a new ERP  $\mathbf{p}$ , the nearest-background distance is  $d_{\mathcal{B}}(\mathbf{p}) = \min_{\mathbf{b} \in \mathcal{B}} \|\mathbf{p} - \mathbf{b}\|_2$ , and only points satisfying  $d_{\mathcal{B}}(\mathbf{p}) > d_{\text{MTI}}$  are kept. The nearest-neighbour search is implemented with a  $k$ -d tree.

*e) DBSCAN clustering.:* A real object generates several nearby ERPs, so the surviving cloud is clustered with DBSCAN [8], which does not require the number of objects in advance, ignores isolated noise, and works well on locally dense point clouds. For each cluster  $\mathcal{C}_k$ , the strongest point is taken as the representative position,  $\mathbf{z}_k = \mathbf{p}_{n^*}$  with  $n^* = \arg \max_{n \in \mathcal{C}_k} \overline{P}_n$ , and the cluster power is  $P_k = \sum_{n \in \mathcal{C}_k} \overline{P}_n$ . Candidates are sorted by  $P_k$  and pruned: if  $\|\mathbf{z}_i - \mathbf{z}_j\|_2 < d_{\text{sep}}$ , the weaker is dropped, so the same physical object is not reported twice. At most  $N_{\max}$  candidates are emitted, each as a tuple  $\mathcal{D}_k = (k, \mathbf{z}_k, P_k)$ .

## G. Multi-Target Tracking Module

Raw detections are noisy and intermittent, so they are stabilised by a constant-velocity Kalman filter with global Hungarian data association [9].

Each target is described by a six-dimensional state  $\mathbf{x}_t = [x_t, y_t, z_t, v_{x,t}, v_{y,t}, v_{z,t}]^\top$  that follows  $\mathbf{x}_{t|t-1} = F\mathbf{x}_{t-1|t-1} + \mathbf{w}_t$  with the constant-velocity transition matrix

$$F = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad (4)$$

and a position-only measurement model  $\mathbf{z}_t = H\mathbf{x}_t + \mathbf{n}_t$  with  $H = [I_3 \quad \mathbf{0}_{3 \times 3}]$ . New tracks are initialised at the detected position with zero velocity. The filter uses  $P_0 = 5I_6$ ,  $Q = 0.1I_6$ , and  $R = 1.0I_3$ , encoding moderate uncertainty in the initial state, limited deviation from constant velocity, and non-negligible sensing noise. The default sampling interval is  $\Delta t = 1.0$ .

At each frame, all active tracks are propagated by  $\hat{\mathbf{x}}_{t|t-1} = F\hat{\mathbf{x}}_{t-1|t-1}$  and  $P_{t|t-1} = FP_{t-1|t-1}F^\top + Q$ . The current detection set is de-duplicated by removing any detection within  $d_{\text{birth}}$  of a stronger one. A cost matrix  $C \in \mathbb{R}^{N_T \times N_D}$  is then built, with entries  $c_{ij} = \|\hat{\mathbf{p}}_{t|t-1}^{(i)} - \mathbf{z}_{t,j}\|_2$ ; entries above the gating threshold  $c_{\max}$  are replaced by a large penalty (1000). The optimal global assignment is obtained by solving

$\pi^* = \arg \min_{\pi} \sum_{(i,j) \in \pi} c_{ij}$  with the Hungarian algorithm. For each accepted pair, a standard Kalman update is performed,

$$\mathbf{y}_t = \mathbf{z}_{t,j} - H\hat{\mathbf{x}}_{t|t-1}, \quad (5)$$

$$S_t = HP_{t|t-1}H^\top + R, \quad (6)$$

$$K_t = P_{t|t-1}H^\top S_t^{-1}, \quad (7)$$

$$\hat{\mathbf{x}}_{t|t} = \hat{\mathbf{x}}_{t|t-1} + K_t\mathbf{y}_t, \quad (8)$$

$$P_{t|t} = (I - K_tH)P_{t|t-1}. \quad (9)$$

Unmatched tracks have their missed-frame counter incremented, so short-term sensing failures (occlusion, weak returns, detector dropout) are tolerated without immediate deletion. A track is deleted when  $m_i \geq M_{\max}$ , and a detection only spawns a new track when its distance to every existing track exceeds  $d_{\text{birth}}$ . A track is exported only after it has reached a minimum age  $A_{\min}$  and accumulated displacement  $\|\hat{\mathbf{p}}_i - \mathbf{p}_{\text{start},i}\|_2 > d_{\min}$ , suppressing short-lived false tracks and clutter.

## H. Sensing-Informed Adaptive Beamforming

Once one or more tracks are confirmed, their estimated positions become steering points for the communication transmission. For each tracked target at  $\mathbf{p}_m$ , the steering vector relative to the gNB is  $\mathbf{u}_m = \mathbf{p}_m - \mathbf{p}_{\text{gNB}}$ , with spherical angles  $\theta_m = \arccos(u_{m,z}/\|\mathbf{u}_m\|_2)$ ,  $\phi_m = \text{atan2}(u_{m,y}, u_{m,x})$ .

Rather than computing a digital precoder, the framework synthesises a custom antenna element pattern with one Gaussian lobe per tracked target. Let  $\hat{\mathbf{r}}(\theta, \phi)$  denote the unit direction toward the observation angle and  $\hat{\mathbf{r}}(\theta_m, \phi_m)$  the steering direction toward target  $m$ . The angular mismatch is  $\psi_m(\theta, \phi) = \arccos(\hat{\mathbf{r}}(\theta, \phi)^\top \hat{\mathbf{r}}(\theta_m, \phi_m))$ . With beamwidth parameter  $\sigma = \text{deg2rad}(W)$ , the multi-lobe gain is

$$G(\theta, \phi) = \sum_{m=1}^M \exp\left(-\frac{\psi_m(\theta, \phi)^2}{2\sigma^2}\right). \quad (10)$$

An array-level gain compensation  $\gamma = 200$  is applied and the pattern is converted to a vertically polarised field amplitude,

$$A_\theta(\theta, \phi) = \sqrt{\max(G(\theta, \phi), 10^{-6})} \gamma, \quad A_\phi(\theta, \phi) = 0. \quad (11)$$

The pattern is regenerated only when the rounded target-position signature changes and a minimum update interval has elapsed, which prevents beam churn under small target motion.

## I. Communication Module

With the updated multi-lobe pattern installed, the path solver executes a ray-tracing sweep to enumerate all viable paths between the transmitter and the UEs. For each viable path, the raw Channel Frequency Response (CFR) is extracted across multiple OFDM subcarriers and decoupled into a scalar pathloss (distance attenuation) and a normalised matrix CFR (spatial/frequency-selective fading), to match ns-3's distinct PropagationLoss and PhasedArray model interfaces. The delay, real and imaginary parts of the normalised CFR, absolute pathloss, and LoS flag are packaged into per-Tx-Rx propagation records and exported to ns-3. To avoid recomputation, ns-3 caches the Sionna response over a short interval (e.g., 80 ms in the warehouse runs).

### J. Implementation

The sensing, tracking, and beamforming modules are implemented in `sionnart.py`, while the mobility module is implemented in `mobility.py`. To interoperate with ns-3, both Python modules are embedded in C++ via `sionna-py-embed.{h,cc}` and `mobility-py-embed.{h,cc}`, which import the Python functions and invoke them on demand. Situation awareness is enabled by setting the `sa_enable` flag at scenario construction; when enabled, an additional query interface `SionnaGetDetectedObjects(double since_time_s)` exposes confirmed ISAC detections to ns-3 application logic. Each detection record contains the simulation timestamp, a stable track identifier, and the estimated 3D world position, allowing IoT applications to consume the gNB's sensing output for situation-aware decision-making.

## IV. EXPERIMENTAL EVALUATION

This section evaluates the proposed framework. Section IV-A defines the scope and methodology. Sections IV-B and IV-C describe the two experiments—a controlled scene-level study and the warehouse Digital-Twin study. Sections IV-D and IV-E present the corresponding results. Section IV-F discusses cross-experiment implications.

### A. Evaluation Scope and Methodology

The evaluation is organised in two complementary stages with a shared measurement logic. The first (Experiment I) is a controlled validation that operates at the SionnaRT level under deterministic trajectories, fixed radio parameters, and three canonical scenes. It isolates the implemented sensing and beamforming logic from upper-layer effects so that scene-driven changes in communication and sensing outputs can be observed cleanly. The second (Experiment II) couples the same framework to a full NR stack and to the warehouse Digital-Twin workload of Section III-A, examining whether the framework remains beneficial under mixed traffic, autonomous mobility, and end-to-end protocol behaviour.

In both stages a baseline (situation-awareness disabled) is compared against ISAC-enabled configurations using a common metric grouping: *communication quality* (path loss, delay, throughput, jitter, packet loss); *sensing quality* (detection rate, false-positive rate, missed-detection rate, mean position error, track continuity, mean track run length); and *system cost* (consumed energy, average power, energy efficiency).

### B. Experiment I: Controlled Validation Setup

Three scenes are used: a free-space scene with the transmitter at  $[0, -10, 10]$  m, a warehouse scene with the same transmitter position, and an urban-micro street-canyon scene with the transmitter at  $[0, -20, 8]$  m (Fig. 4). The free-space and warehouse scenes use the `iw_hub` mobile-robot mesh as the target object; the street-canyon scene uses a car mesh. Three targets follow deterministic planar trajectories—sinusoidal in the indoor scenes, lane-aligned in the street-canyon—over 100 time steps at 0.2 s resolution, for a total observation window of 20 s.

Fig. 4. Controlled-validation scenes: free space, warehouse, and urban micro street canyon.

TABLE I  
EXPERIMENT I RADIO AND SENSING PARAMETERS.

| Parameter                         | Value                            |
|-----------------------------------|----------------------------------|
| Carrier frequency                 | 3.5 GHz                          |
| Subcarrier spacing                | 30 kHz                           |
| Number of subcarriers             | 648                              |
| Transmission power                | 46 dBm                           |
| Tx array                          | $8 \times 8$ planar              |
| Rx array                          | $2 \times 2$ planar              |
| Antenna pattern                   | TR 38.901, vertical polarisation |
| Min sensing power $P_{\min}$      | $10^{-25}$                       |
| Clustering $\varepsilon$ (DBSCAN) | 1.2                              |
| Min displacement $d_{\min}$       | 0.15 m                           |

The shared radio parameters are listed in Table I.

Seven configurations are compared. `baseline` disables situation awareness, MTI, and tracking. `isac_full` enables all three with beamwidth  $W = 18^\circ$  and propagation depth 3. `isac_no_mti` disables MTI; `isac_narrow` ( $W = 12^\circ$ ) and `isac_wide` ( $W = 30^\circ$ ) sweep beamwidth; `isac_deep` raises propagation depth to 5; `isac_no_tracker` disables the tracker. At each step, estimated track positions are matched to ground truth with the Hungarian algorithm; unmatched ground-truth targets count as missed detections, unmatched estimates as false positives, and matched-pair distances as localisation errors.

### C. Experiment II: Warehouse Use-Case Setup

The warehouse experiment is implemented as two ns-3 scenarios, `warehouse-sionna` (baseline) and `warehouse-isac` (situation-awareness enabled), which share the same topology, radio configuration, mobility model, and trace-export logic and differ only in the Sionna backend initialisation. Both scenarios use the same warehouse XML scene for ray tracing and the same OBJ for PyBullet mobility. Sionna-backed channel helper, proportional-fair OFDMA scheduling, rank-limited MIMO PMI, HARQ retransmissions, and a fixed gNB slot pattern are common to both scenarios.

The shared radio and Sionna parameters are listed in Table II. The wireless and wired node placements appear in Tables III and IV; the application workload appears in Table V.

For each gNB array size in  $\{2 \times 2, 4 \times 4, 8 \times 8\}$ , both scenarios export traces for application flows, propagation, mobility,

TABLE II  
EXPERIMENT II RADIO AND SENSING PARAMETERS.

| Parameter                         | Value                                      |
|-----------------------------------|--------------------------------------------|
| Carrier frequency                 | 3.7 GHz                                    |
| Subcarrier spacing                | 30 kHz                                     |
| Number of subcarriers             | 648                                        |
| gNB Tx power                      | 14 dBm                                     |
| Mobility update interval          | 0.05 s                                     |
| Propagation cache duration        | 0.08 s                                     |
| gNB array sizes (swept)           | $2 \times 2$ , $4 \times 4$ , $8 \times 8$ |
| UE array size                     | $2 \times 2$                               |
| Maximum propagation depth         | 4                                          |
| Beamwidth $W$                     | $10^\circ$                                 |
| Min sensing power $P_{\min}$      | $10^{-25}$                                 |
| Clustering $\varepsilon$ (DBSCAN) | 0.5                                        |
| Min displacement $d_{\min}$       | 0.15 m                                     |

TABLE III  
WIRELESS NODE PLACEMENT IN EXPERIMENT II.

| Node type          | Count | Position(s) [m]                              |
|--------------------|-------|----------------------------------------------|
| gNB                | 1     | (14.0, -11.0, 6.5)                           |
| Temperature sensor | 1     | (10.2, 9.0, 4.5)                             |
| Humidity sensor    | 1     | (11.5, 8.5, 4.5)                             |
| Fixed cameras      | 2     | (6.5, 12.0, 3.0), (-13.0, -11.0, 3.0)        |
| Package sensors    | 3     | (9, 12, 0.2), (9, 7, 0.2), (9, 2, 0.2)       |
| Rack sensors       | 3     | (-12, -9, 0.2), (-7, -6, 0.2), (-2, -4, 0.2) |
| Mobile robots      | 3     | (9, 10, 0), (9, 5, 0), (9, 0, 0)             |

MQTT timing, energy consumption, and a system summary; the ISAC scenario additionally exports detected-object records (estimated position, ground-truth robot position, geometric error). Application metrics are aggregated over included wireless flows (MQTT and Video), with throughput summed and delay/jitter computed as received-packet-weighted means.

#### D. Experiment I: Controlled-Validation Results

1) *Free-space scene*: The free-space results are summarised in Tables VI and VII. Relative to the baseline, all ISAC variants reduce mean path loss by 15.6–18.0 dB, with corresponding gains in received power, while mean delay remains constant at 54.10 ns—signal strength dominates the change. Among the ablations, *isac\_wide* gives the lowest mean path loss (84.39 dB).

The tracked variants achieve detection rates in [0.847, 0.903] at very low false-positive rates ( $\approx 0.018$ – $0.019$ ) and tightly grouped position errors (1.12–1.15 m). The *isac\_no\_tracker* variant raises raw detection rate to 0.970 but increases the false-positive rate to 0.176, showing that removing the tracker improves raw sensitivity at the cost of output stability. Free space confirms the expected upper-bound behaviour of the framework.

2) *Warehouse scene*: Tables VIII and IX report the warehouse-scene results. Path loss reductions are smaller than in free space (7.4–14.7 dB) but still consistent; *isac\_wide* again yields the strongest communication. Sensing, however, degrades sharply: detection rate plateaus around 0.28–0.29 with missed-detection rate near 0.71 and track continuity around 0.27–0.28. Position error stays close to 1.0–1.13 m for the tracked variants, so localisation is reasonably accurate *when* an object is detected, but the dominant limitation is the frequency of missed detections. *isac\_deep*

TABLE IV  
WIRED NODE PLACEMENT IN EXPERIMENT II.

| Node type     | Count | Position(s) [m]                                                       |
|---------------|-------|-----------------------------------------------------------------------|
| Video clients | 5     | (-4, 8, 1.5), (-3, 3, 1.5), (-6, 10, 1.5), (-6, 6, 1.5), (-6, 1, 1.5) |
| Edge node     | 1     | (0, 0, 1.5)                                                           |
| Remote host   | 1     | (-5, 5, 1.5)                                                          |

TABLE V  
APPLICATION WORKLOAD IN EXPERIMENT II.

| Application / process         | Setting         |
|-------------------------------|-----------------|
| Temperature publish interval  | 250 ms          |
| Humidity publish interval     | 250 ms          |
| Package-sensor check interval | 40 ms           |
| Withdrawal check interval     | 250 ms          |
| Withdrawal probability        | 0.85            |
| Robot telemetry interval      | 10 ms           |
| Robot-mounted camera          | 8 fps, 32,768 B |
| Fixed camera                  | 6 fps, 24,576 B |

and *isac\_no\_tracker* produce substantially higher false-positive rates, indicating that deeper propagation modelling or removal of tracking worsens stability indoors.

3) *Urban-micro street-canyon scene*: In the street canyon, communication metrics are essentially invariant across configurations (Table X): path loss  $\approx 72.83$  dB, delay 44.84 ns, and Rx power  $\approx -26.83$  dBm in all cases except *isac\_deep*, which marginally reduces path loss to 72.70 dB. The dominant geometric structure of the canyon governs communication more strongly than the tested ISAC ablations. Sensing (Table XI) is moderate: the best tracked variants reach detection rate 0.437 at false-positive rates of 0.127–0.138, but track continuity remains low (0.187). *isac\_full* and *isac\_deep* produce higher false-positive rates; *isac\_no\_tracker* collapses both detection rate and continuity. Although *isac\_no\_tracker* attains the lowest mean position error, the small set of matched detections prevents this from indicating better overall sensing.

4) *Cross-scene summary*: Free space provides the strongest joint sensing-and-communication performance and confirms the framework's expected upper bound. The warehouse scene preserves communication gains but exposes a strong sensing penalty from indoor clutter. The street canyon shows that, once a confined geometry dominates propagation, communication becomes nearly invariant to ISAC ablations while sensing remains configuration-sensitive. Across all three scenes, the tracked variants consistently produce cleaner outputs than *isac\_no\_tracker*, confirming that temporal association is required for stable ISAC behaviour.

#### E. Experiment II: Warehouse Use-Case Results

1) *Application-level KPIs*: The application-layer comparison (Fig. 5) shows that the main differences between baseline and ISAC are concentrated on the controller, package-sensor, and robot traffic. At  $2 \times 2$ , controller throughput drops from 417.674 to 205.345 kbps under ISAC, and package-sensor throughput from 22.990 to 6.634 kbps; robot throughput slightly increases from 1159.444 to 1249.326 kbps but is accompanied by 15.439% packet loss, so the  $2 \times 2$  ISAC

TABLE VI  
FREE-SPACE SCENE—PROPAGATION METRICS.

| Variant         | Path loss [dB] | Delay [ns] | Rx power [dBm] |
|-----------------|----------------|------------|----------------|
| baseline        | 102.39         | 54.10      | −56.39         |
| isac_full       | 86.30          | 54.10      | −40.30         |
| isac_no_mti     | 85.53          | 54.10      | −39.53         |
| isac_narrow     | 86.82          | 54.10      | −40.82         |
| isac_wide       | 84.39          | 54.10      | −38.39         |
| isac_deep       | 85.54          | 54.10      | −39.54         |
| isac_no_tracker | 85.43          | 54.10      | −39.43         |

TABLE VII  
FREE-SPACE SCENE—SENSING METRICS.

| Variant         | Det. rate | FPR   | Miss  | Err. [m] | Cont. |
|-----------------|-----------|-------|-------|----------|-------|
| baseline        | 0.000     | 0.000 | 1.000 | —        | 0.000 |
| isac_full       | 0.847     | 0.019 | 0.153 | 1.148    | 0.550 |
| isac_no_mti     | 0.897     | 0.018 | 0.103 | 1.125    | 0.537 |
| isac_narrow     | 0.897     | 0.018 | 0.103 | 1.125    | 0.537 |
| isac_wide       | 0.897     | 0.018 | 0.103 | 1.125    | 0.537 |
| isac_deep       | 0.903     | 0.018 | 0.097 | 1.126    | 0.510 |
| isac_no_tracker | 0.970     | 0.176 | 0.030 | 1.208    | 0.583 |

configuration is characterised by unstable delivery. At  $4 \times 4$ , the controller and package-sensor throughputs return to baseline levels and robot throughput rises to 1582.081 kbps versus a baseline of 1328.452 kbps. The same robot-dominant gain persists at  $8 \times 8$  (1445.378 kbps), without proportional additional benefit. The principal communication gain of ISAC in the warehouse therefore appears once the gNB array reaches  $4 \times 4$ .

2) *Propagation KPIs*: Fig. 6 explains the throughput trends. Robot path loss in the baseline drops from 83.948 dB at  $2 \times 2$  to roughly 69.2 dB at  $4 \times 4$  and  $8 \times 8$ ; under ISAC the same trend is stronger, falling from 82.318 dB at  $2 \times 2$  to 57.359 dB at  $4 \times 4$  and remaining essentially unchanged at  $8 \times 8$ . The rack-sensor links also benefit, with about a 20 dB reduction relative to the baseline at all three array sizes. By contrast, the package, temperature, and humidity sensors all see higher path loss under ISAC. The propagation improvement is therefore selective: it is concentrated on the links that are most relevant for mobile warehouse operation (the robots) and this selective gain drives the system-level throughput improvement at  $4 \times 4$  and beyond.

3) *Energy KPIs*: The principal scaling factor for power consumption is the gNB array size: gNB average power rises from  $\approx 25.6$  W at  $2 \times 2$  to  $\approx 51.3$  W at  $4 \times 4$  and  $\approx 145$  W at  $8 \times 8$ , with corresponding gNB energy of  $\approx 3.1$ ,  $\approx 6.15$ , and  $\approx 17.4$  kJ. UE-side power is approximately invariant across array sizes because all UEs share a  $2 \times 2$  antenna configuration. At the system level, total reported energy is nearly identical between baseline and ISAC for the  $4 \times 4$  and  $8 \times 8$  cases while ISAC achieves slightly higher throughput, so the  $4 \times 4$  ISAC configuration improves end-to-end performance without significant additional system-level energy cost.

4) *Sensing and Localisation*: Fig. 8 reports the warehouse sensing KPIs derived from active robot opportunities and validated detected-object records. Detection rate rises from 0.485 at  $2 \times 2$  to 0.523 at  $4 \times 4$ , with  $8 \times 8$  essentially unchanged at 0.526; missed-detection rate correspondingly falls from 0.515 to  $\approx 0.477$ . Validated false positives are

TABLE VIII  
WAREHOUSE SCENE—PROPAGATION METRICS.

| Variant         | Path loss [dB] | Delay [ns] | Rx power [dBm] |
|-----------------|----------------|------------|----------------|
| baseline        | 109.16         | 53.95      | −63.16         |
| isac_full       | 97.90          | 53.95      | −51.90         |
| isac_no_mti     | 97.39          | 53.95      | −51.39         |
| isac_narrow     | 101.75         | 53.95      | −55.75         |
| isac_wide       | 94.50          | 53.95      | −48.50         |
| isac_deep       | 97.44          | 53.95      | −51.44         |
| isac_no_tracker | 98.52          | 53.95      | −52.52         |

TABLE IX  
WAREHOUSE SCENE—SENSING METRICS.

| Variant         | Det. rate | FPR   | Miss  | Err. [m] | Cont. |
|-----------------|-----------|-------|-------|----------|-------|
| baseline        | 0.000     | 0.000 | 1.000 | —        | 0.000 |
| isac_full       | 0.280     | 0.023 | 0.720 | 1.032    | 0.273 |
| isac_no_mti     | 0.290     | 0.033 | 0.710 | 1.011    | 0.283 |
| isac_narrow     | 0.290     | 0.033 | 0.710 | 1.012    | 0.283 |
| isac_wide       | 0.290     | 0.022 | 0.710 | 1.012    | 0.283 |
| isac_deep       | 0.290     | 0.121 | 0.710 | 1.129    | 0.283 |
| isac_no_tracker | 0.283     | 0.336 | 0.717 | 1.252    | 0.273 |

already low at  $2 \times 2$  (0.021) and drop to zero for  $4 \times 4$  and  $8 \times 8$ . Track continuity remains close to unity across all three configurations, indicating that once a valid detection sequence is formed the exported track identities are stable.

The localisation-error distribution (Fig. 9) shows that median accuracy is not the principal differentiator:  $2 \times 2$  has the lowest median error (0.384 m) but the heaviest upper tail (max 10.427 m), whereas  $4 \times 4$  and  $8 \times 8$  have slightly larger medians (0.577 m) but significantly lower extreme errors ( $\approx 4.5$  m max) and lower P90/P95. The larger arrays therefore improve sensing *stability*—suppressing large outliers and sustaining more valid matched opportunities—rather than pointwise median precision.

## F. Discussion

The two experiments show that the proposed ISAC framework behaves consistently but in a strongly condition-dependent manner. Under controlled conditions, the framework responds to scene geometry and ablation choices in a physically meaningful way: free space yields the most favourable regime with strong communication gains and stable sensing when the tracker is retained; the warehouse scene preserves communication gains but introduces a substantial sensing penalty in completeness and continuity; and the urban micro scene shows that strongly geometry-driven propagation can decouple communication stability from sensing stability.

The warehouse Digital-Twin study extends these observations into a deployment-relevant context. Practical applicability depends not only on whether ISAC is enabled but also on whether the radio configuration provides sufficient operating margin. The  $2 \times 2$  ISAC configuration fails to sustain competitive end-to-end communication, while the  $4 \times 4$  and  $8 \times 8$  configurations restore system-level competitiveness and slightly exceed the baseline in throughput and efficiency. The  $8 \times 8$  configuration does not materially improve upon  $4 \times 4$  in either communication or sensing while doubling and tripling the gNB power draw, identifying  $4 \times 4$  as the balanced operating point for the tested workload.

TABLE X  
URBAN-MICRO SCENE—PROPAGATION METRICS.

| Variant         | Path loss [dB] | Delay [ns] | Rx power [dBm] |
|-----------------|----------------|------------|----------------|
| baseline        | 72.83          | 44.84      | −26.83         |
| isac_full       | 72.83          | 44.84      | −26.83         |
| isac_no_mti     | 72.83          | 44.84      | −26.83         |
| isac_narrow     | 72.83          | 44.84      | −26.83         |
| isac_wide       | 72.83          | 44.84      | −26.83         |
| isac_deep       | 72.70          | 44.84      | −26.70         |
| isac_no_tracker | 72.83          | 44.84      | −26.83         |

TABLE XI  
URBAN-MICRO SCENE—SENSING METRICS.

| Variant         | Det. rate | FPR   | Miss  | Err. [m] | Cont. |
|-----------------|-----------|-------|-------|----------|-------|
| baseline        | 0.000     | 0.000 | 1.000 | —        | 0.000 |
| isac_full       | 0.400     | 0.310 | 0.600 | 0.933    | 0.177 |
| isac_no_mti     | 0.437     | 0.127 | 0.563 | 0.972    | 0.187 |
| isac_narrow     | 0.437     | 0.138 | 0.563 | 0.972    | 0.187 |
| isac_wide       | 0.437     | 0.138 | 0.563 | 0.972    | 0.187 |
| isac_deep       | 0.380     | 0.367 | 0.620 | 0.963    | 0.177 |
| isac_no_tracker | 0.183     | 0.329 | 0.817 | 0.686    | 0.020 |

The sensing analysis reinforces this conclusion: larger gNB arrays do not primarily reduce median localisation error, but they suppress large-error outliers and sustain a larger pool of valid matched opportunities. In deployment terms, this is the form of stability that is most useful, because isolated low-error detections accompanied by frequent misses or unstable tracks are of limited practical value.

At the same time, the results reveal residual constraints. Sensing is significantly more fragile than communication in cluttered environments; communication gains are not uniform across device classes, so application-level benefits remain workload-dependent; and the warehouse sensing evaluation is currently derived from exported detections and reconstructed opportunity definitions rather than from a complete tracker-side confusion matrix. These observations frame the future-work agenda discussed in Section V.

## V. CONCLUSION AND FUTURE WORK

This paper presented a situation-aware Integrated Sensing and Communication (ISAC) framework that couples NVIDIA Sionna RT with ns-3/5G LENA into a single co-simulation pipeline, and evaluated it through two complementary studies. A controlled validation across free-space, warehouse, and urban-micro street-canyon scenes confirmed that the framework responds to scene geometry and configuration changes in a physically meaningful way: it exhibits the expected upper-bound behaviour in free space, preserves communication gains while degrading sensing in cluttered indoor settings, and decouples communication stability from sensing stability under structured outdoor multipath. A warehouse Digital-Twin study under three gNB array sizes then showed that the same framework remains beneficial in a realistic deployment-oriented setting once sufficient array support is available. The  $4 \times 4$  configuration emerged as the balanced operating point: it improves the dominant robot-related communication and sensing behaviour without incurring the substantially higher gNB power required by the  $8 \times 8$  configuration.

Fig. 5. Per-node-type application metrics under baseline and ISAC for  $2 \times 2$ ,  $4 \times 4$ , and  $8 \times 8$  gNB arrays.

Fig. 6. Per-node-type propagation metrics (path loss and path delay).

Several limitations remain. First, sensing robustness degrades in cluttered environments, particularly with respect to missed detections and track continuity. Second, the communication benefit of ISAC is not uniform across device classes, so application-level gains remain partly workload-dependent. Third, the warehouse sensing evaluation is derived from exported detections and reconstructed opportunity definitions rather than from a complete tracker-side confusion matrix. These limitations do not invalidate the present results, but they delimit the claims supported by the current implementation.

Future work will proceed along three directions. The first is algorithmic refinement of the sensing and tracking pipeline for cluttered environments, including more robust association, filtering, and false-track suppression. The second is an expanded experimental campaign with additional seeds, traffic regimes, and scene conditions, so that the conclusions can be supported by broader statistical evidence. The third is deeper optimisation of the communication–sensing control logic, particularly for beam management. The present evaluation indicates that position-based beamforming alone is not sufficient to guarantee the best operating point across all conditions; a more effective approach is expected to combine position information with channel state information (CSI) so that beam selection exploits both geometric awareness and instantaneous channel realisations. Together, these steps would extend the framework

Fig. 7. Average power and consumed energy per node group under baseline and ISAC.

Fig. 9. Localisation-error ECDF and per-configuration distribution.

Fig. 8. Warehouse sensing KPIs for  $2 \times 2$ ,  $4 \times 4$ , and  $8 \times 8$  gNB arrays.

from a proof of viability toward a deployment-ready ISAC evaluation environment.

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